Green Chemistry

PAPER

View Article Online
View Journal | View Issue

Cite this: *Green Chem.*, 2023, **25**, 10061

A multifunctional lignin-based composite ultra-adhesive for wood processing†

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Received 20th September 2023,
Accepted 6th November 2023

DOI: 10.1039/d3gc03542e

rsc.li/greenchem

Phenolic resin adhesives are usually synthesized using phenol and formaldehyde. Both of them are fossil-based and the resins are not biodegradable. Replacing them with whole biomass-based lignin and furfural is attractive. However, it is still difficult to wholly replace them, especially lignin due to its low reactivity. In this paper, a preparation method for a composite adhesive was presented. A lignin-based composite adhesive was prepared by using a deep eutectic solvent (DES), lignin, and furfural. The lignin-based composite adhesive is a machinable adhesive for wood, and the adhesive strength for the paulownia plate reached 5.71 MPa. The adhesive also possessed outstanding flame retardance performance with a V0 grade and the ultimate oxygen index of the adhesive-coated wood was 30%. Interestingly, it also had excellent thermal insulation and photothermal performances. Importantly, wood chips can also be processed into strong regenerated wood using the adhesive.

Introduction

Phenolic resin adhesives have been developed since 1911 and rely mainly on phenol and formaldehyde as feedstocks.[1–3] They exhibit excellent performance as adhesives for wood, and they are widely used in various engineered wood products.[4,5] However, phenol and formaldehyde are mainly derived from fossil resources such as petroleum. With the development of a sustainable society, the demand for green adhesives is gradually rising. In addition, the adhesives prepared from phenol and formaldehyde are also not biodegradable.

The use of low-toxicity or non-toxic biomass feedstocks instead of phenol and formaldehyde is an important approach for the preparation of more environmentally friendly adhesives. As one of the most abundant natural aromatic polymers and the only non-petroleum source of naturally renewable aromatic compounds, lignin is a satisfactory substitute for phenol because of the presence of benzene propane units and abundant phenolic hydroxyl groups, which are structurally similar to phenol.[6–11] Therefore, efforts have been made to use lignin as an alternative to phenol in adhesives.[12,13] In addition, formaldehyde is a bulk chemical used in wood products which results in plenty of environmental problems due to its high volatility and toxicity to humans, which will be replaced by green aldehydes. Furfural, which is less toxic compared to formaldehyde, can be obtained in mass from the degradation of hemicellulose of biomass feedstock and can serve as an excellent substitute for formaldehyde.[14–16]

However, lignin has low reactivity and fewer reaction sites with aldehydes compared to phenol, which is due to the following four reasons:[17,18] (I) the methoxy group is located at the *ortho* position of the phenolic hydroxyl group in the guaiacyl and syringyl monomers; (II) the connection between units (such as 5-5, α-O-5, β-5 connection) is located at the *para* position of the phenolic hydroxyl group; (III) the phenolic hydroxyl groups participate in the connection between lignin monomers (such as β-O-4 bonds); (IV) the high-order structure of lignin leads to large steric hindrance.[19,20] To address these problems, various modifications of lignin have been carried out to increase its reactive sites, and the commonly used modification methods reported so far include phenolation, methylolation, and demethylation.[21–25] However, these methods not only require the use of petroleum-based compounds or toxic and hazardous substances, but also add additional processing steps and costs.[19]

In addition, it was found that the reactive sites of lignin can be increased by depolymerizing lignin.[26–32] Plenty of reports indicate that ionic liquids can depolymerize lignin well, but ionic liquids are costly with unclear toxicity.[33–36] Deep eutectic solvents (DESs) are multi-component mixtures consisting of a certain molar ratio of hydrogen bond acceptors (*e.g.* quaternary ammonium salts) and hydrogen bond donors (*e.g.* compounds such as amides, carboxylic acids, and polyols), whose melting point is significantly lower than that of each component.[37–42] Most importantly, DESs are also good solvents for lignin, making it possible to homogeneously react or process lignin.[43]

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† Electronic supplementary information (ESI) available. See DOI: **https://doi.org/10.1039/d3gc03542e**

The currently reported methods for using acidic DESs have been used for lignin depolymerization. However, the amount of DESs used is too much, which reduces the viscosity of the reaction system and makes the reaction system not easy to form a crosslinked structure.[44,45] However, we believe that alkaline DESs can not only depolymerize lignin, but also make lignin react fully with furfural.[36]

Here, we have achieved the replacement of 100% phenol with lignin and 100% formaldehyde with furfural and have prepared a lignin-based composite adhesive in a DES for the first time using a one-pot method, making the system greener. The DES makes a lignin-based composite adhesive that has excellent adhesive strength towards wood, even at low temperatures (Scheme 1). Our lignin-based composite adhesives have superior properties to lignin adhesives that are currently available. The lignin adhesive prepared by Yang[46] is also a biomass adhesive with many excellent properties, but it has lower binding strength, whereas the DES-lignin-furfural lignin-based composite adhesive prepared by us has even better binding strength.

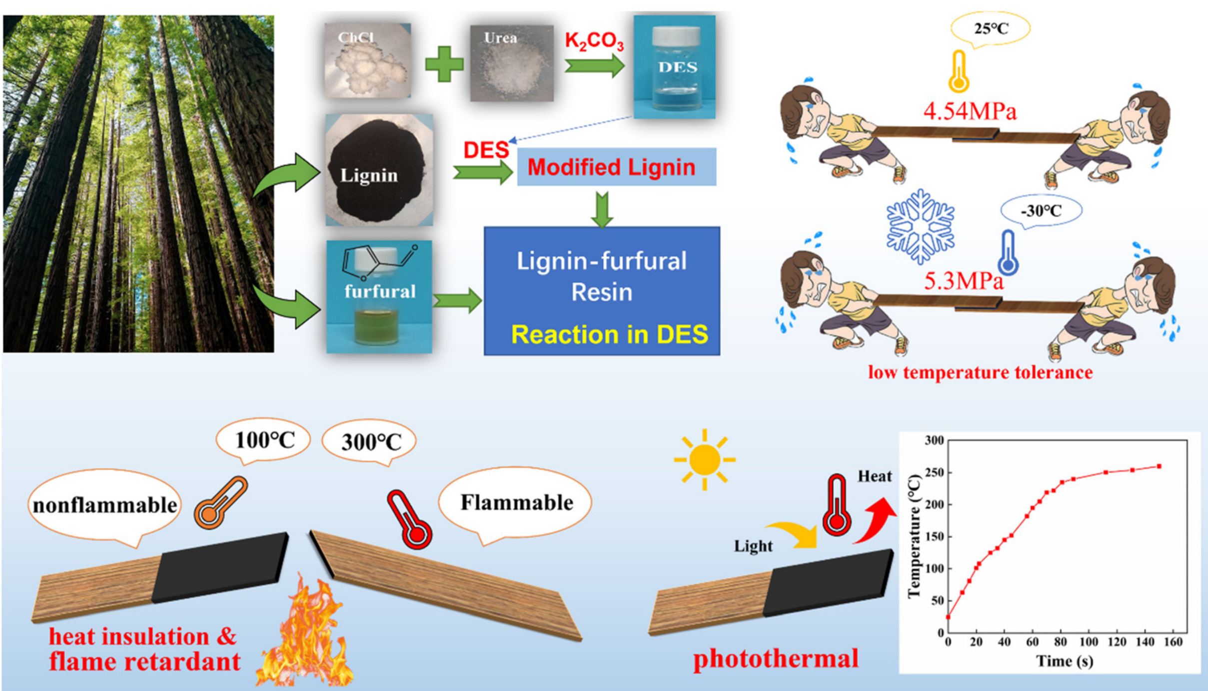
Results and discussion

Lignin-based composite adhesive synthesis in a DES

DES (ChCl/urea with K_2CO_3) and alkaline lignin were first mixed in the mass ratios of 1 : 1, 2 : 1, and 5 : 1, followed by the steps of depolymerization and preparation of the lignin-based composite adhesive. Finally, the adhesive was applied to the paulownia plate, and then hot-pressed for 10 min at 0.13 MPa and 130 °C and cold-pressed for 12 h. We chose the test methods of double-layer shear specimen and three-layer shear specimen because the wood of the three-layer shear specimen was thin and the adhesive bond strength was high; the phenomenon of wood fracture occurred during the test process, so we used the double-layer shear specimen for testing (GB/T 33333-2016) (Fig. S1†).

In Fig. 1, it can be seen that when the mass ratio of DES to lignin was 2 : 1 and the reaction time was 6 h, the adhesion performance could reach 4.54 MPa (Fig. 1b). When the ratio of DES to lignin was 1 : 1, the adhesion performance reduced (2.81 MPa at 3 h of reaction) because the system was too viscous and difficult to stir, which may make the lignin and furfural not in full contact with the reaction, and the amount of DES was too low to fully depolymerize the lignin (Fig. 1a). When the ratio of DES to lignin was 5 : 1, the maximum adhesion performance was only 0.34 MPa and could not be put to use because the high content of the DES made the viscosity of the system too low, failing to cure (Fig. 1c). In summary, the optimal reaction conditions were 6 h at the mass ratio of 2 : 1 (DES : lignin).

The lignin–furfural lignin-based composite adhesive obtained from the reaction for 6 h was applied to the paulownia plate for the adhesion performance test, and the results obtained are shown in Fig. 1d. The as-prepared resin between two wood sheets was hot pressed at 0.13 MPa, and the adhesive strength measured at room temperature was 4.54 MPa (wood 0.13/rt), which far exceeded the 0.7 MPa required by the Chinese National Standard (GB/T 14732-2017), while the adhesive strength of the hot-pressed sample was tested



Scheme 1 Synthesis of a lignin-based composite adhesive in a DES and its excellent performance.

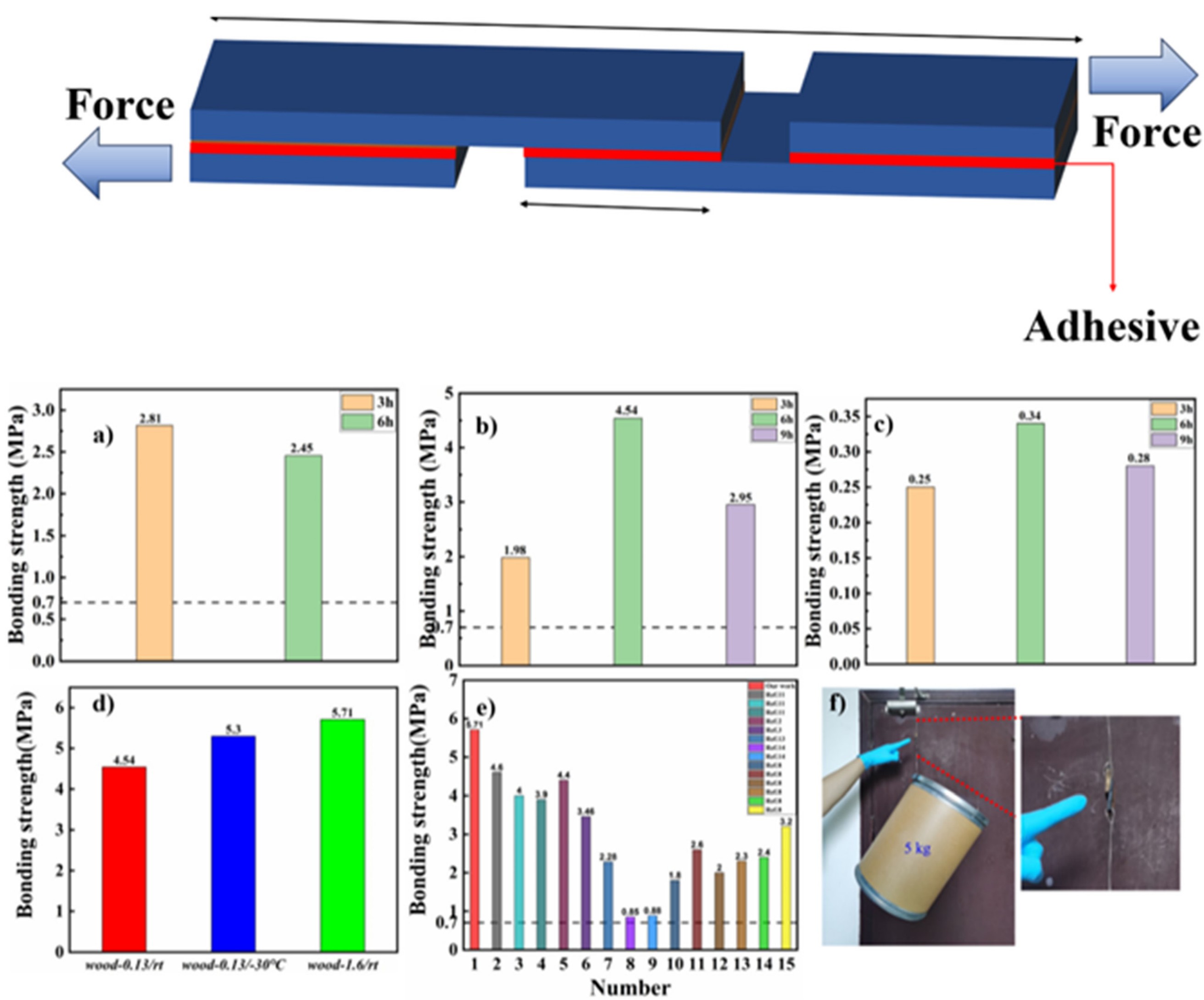


Fig. 1 Adhesion properties at different reaction times when the ratio of the DES to lignin mass was 1 : 1 (a), 2 : 1 (b), and 5 : 1 (c); the adhesive strength of the paulownia plate under different treatment conditions (d), the adhesive strength of our work compared with the currently reported lignin-based composite adhesive (e), and a physical picture of the adhesive bonding of wood panel (f).

immediately after it was placed at −30 °C for 12 h. It was found that the adhesive strength reached 5.3 MPa (wood 0.13/−30 °C), which was a 16.74% increase in adhesive strength compared to leaving it at room temperature for 12 h. The reason may be that the molecular chain spacing became smaller at low temperatures, the molecular chains were arranged more densely and the free volume was reduced. Therefore, the strength of chemical bonds was higher at low temperatures than at room temperature.[47–49] Most of the currently reported phenolic resin wood adhesives were hot pressed at 1.2–1.8 MPa to treat the samples. In contrast, the lignin-based composite adhesive only required hot pressing at 0.13 MPa to achieve comparable or even better results. When hot pressing increased to 1.6 MPa, the adhesive strength at room temperature reached 5.71 MPa (wood 0.13/rt), which basically exceeded most of the currently reported wood adhesives. The reason may be the presence of the DES in the system. The DES in DLF-6 adhesive had a corrosive effect on the wood sheet under hot pressure, destroying the complex structure of the wood by dissolving and modifying lignin *in situ*. Therefore, the adhesive could penetrate more deeply into the wood pieces and thus had higher adhesive strength. In contrast, few of the phenolic resin and urea-formaldehyde resin adhesives reported so far have introduced a DES. The few systems that introduced a DES used it only to depolymerize lignin and then the DES was removed. Therefore, a major innovation of this work was the introduction of a small amount of DES to modify the lignin rather than dissolve it. The results revealed that retaining the right amount of DES had a positive effect on the adhesive strength of the adhesive.

As can be seen from Table S1† and Fig. 1e, the DLF-6 adhesive was not only stronger in bonding, but also greener relative to the currently reported phenolic resin and urea-formaldehyde resin adhesives. Besides, our reaction process was conducted by the one-pot method, and there was no formaldehyde and phenol in the new system. Although some of the currently reported phenolic resins also used other aldehydes to replace formaldehyde, they still used formaldehyde to modify the lignin and did not achieve the true lignin-based raw material preparation of the adhesive.

The viscosity of the adhesives and lignin depolymerization

As shown in Fig. 2a, the C–H vibrational peak of furfural at 3136 cm^{-1}, 2855 cm^{-1}, 2811 cm^{-1}, 1567 cm^{-1}, and 1392 cm^{-1} disappeared in DLF-6 (Table S2†), indicating that there was no remaining furfural in DLF-6, and furfural had been completely disappeared after 6 h of reaction. The characteristic peak of the DES at 2165 cm^{-1} appeared in both D-L and DLF-6, indicating the presence of the DES in them (Fig. 2b). The peaks of DLF basically did not change with the increase in reaction time, indicating that no new functional groups were generated with the increase in reaction time (Fig. 2c). Fig. 2d shows the phenolic hydroxyl peak of A-L at 3440 cm^{-1} due to its large space steric hindrance, and the phenolic hydroxyl peak of D-L shifted to a lower wavenumber, indicating the decrease of the space steric hindrance. Moreover, some new hydroxyl peaks were found in D-L compared with A-L due to the generation of alcoholic hydroxyl groups in D-L, indicating a variety of pheno-

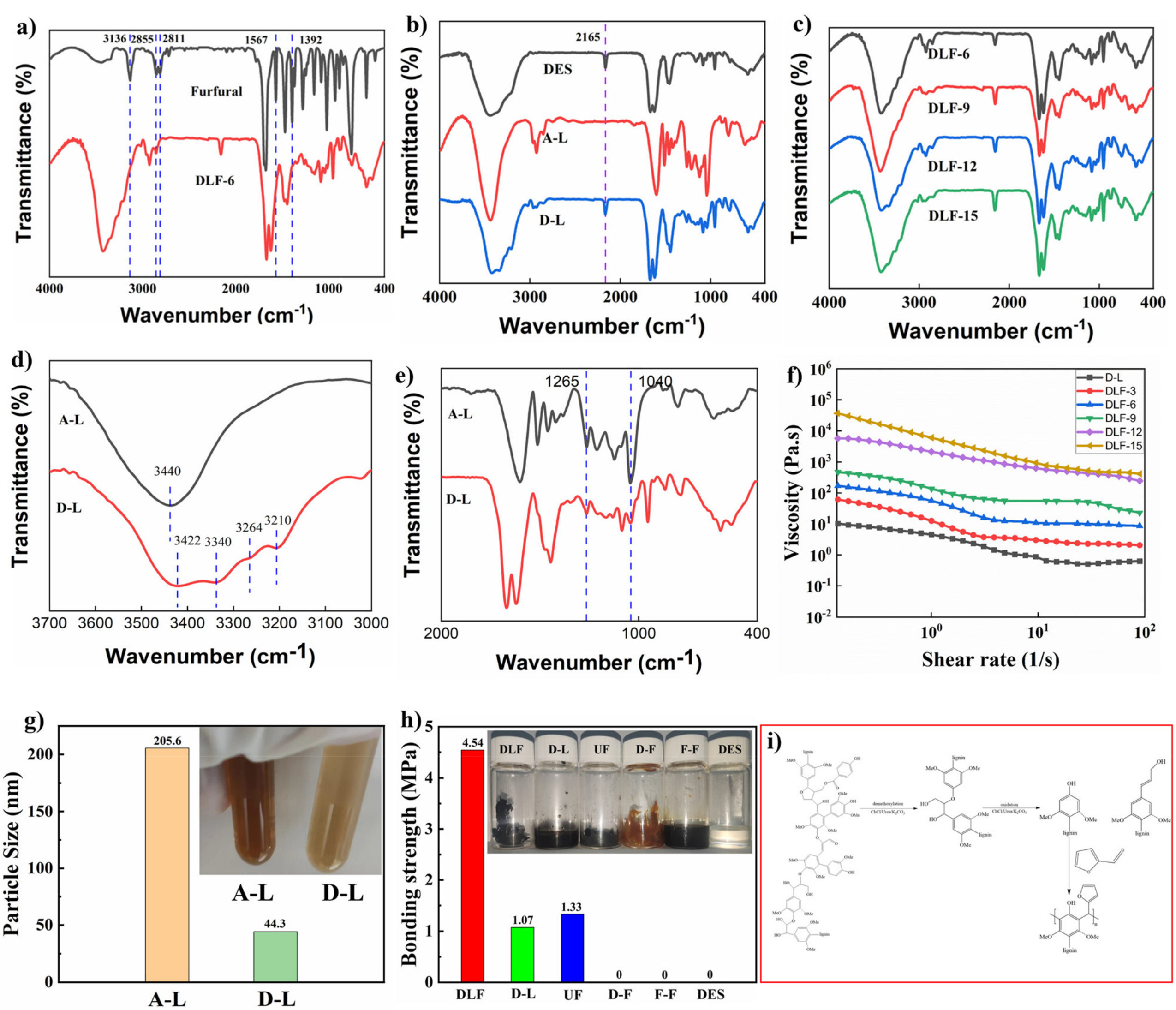


Fig. 2 FTIR spectra of furfural, alkali lignin (A-L), the deep eutectic solvent (DES), the alkali lignin depolymerized by the deep eutectic solvent (D-L), and the lignin-based composite adhesive prepared at different reaction times (DLF-6, 9, 12, 15) (a–c), FIIR spectra of A-L (e) and D-L (d and e), the viscosity of lignin (D-L) depolymerized by the DES and the variation curve of viscosity of the lignin-based composite adhesive (DLF) with the shear rate at different reaction times (f), average particle size tests using DLS for A-L and D-L (g, the inset shows the photographs of A-L and D-L in water), comparison of DLF, UF and D-L bond strengths (h), and the possible degradation mechanism of lignin in the DES and the principle of adhesive synthesis (i).

lic hydroxyl groups with different space steric hindrances. These results revealed that the alkali lignin had been depolymerized. Fig. 2e shows that the peak of the C–O–C group (1265–1040 cm^{-1}) of D-L became weaker relative to that of A-L, which also proved that the alkali lignin in D-L was depolymerized, and the ether bond was broken.

Rheological properties of the lignin (D-L) depolymerized by the DES and the DES-lignin-furfural lignin-based composite adhesive (DLF-3, 6, 9, 12,15) reacted for 3, 6, 9, 12, and 15 h are shown in Fig. 2f. From DLF-3 to DLF-15, the increase in viscosity was caused by the increase in the degree of internal cross-linking of the adhesive in this system with the reaction time. For pure D-L in the DES, the viscosity was the lowest compared with DLFs. The viscosity of both D-L and DLF decreased with increasing shear rate, exhibiting a significant shear thinning property. Moreover, it was evident from their moduli that both their loss modulus and energy storage modulus increased with the reaction time, with the smallest for D-L and the largest for DLF-15. In addition, their loss modulus was larger than the energy storage modulus, indicating that they all became liquid at 130 °C (Fig. S2†).

Fig. 2g shows the results of the DLS studies, and the average particle size of A-L was 205.6 nm while that of D-L was 44.3 nm; and the photo images of the dispersion of A-L and D-L in water are also shown in the inset, and the color of the DES-treated lignin became weak. It revealed that the particle size of alkali lignin became smaller during the depolymerization process, and alkali lignin could be well depolymerized in the DES.

Since the DES contained urea, we guessed that the following reactions may occur in the composite adhesive. First, urea may react with furfural to produce urea-formaldehyde resin. Second, furfural may undergo self-condensation. Thus, we conducted controlled experiments to verify the reaction mechanism. We divided the experiment into five groups: furfural, urea + furfural, DES + furfural, DES + lignin, and DES (Fig. 2h). The proportions and conditions of this experiment were the same as those used for the preparation of the lignin-based composite adhesive. The experimental conclusions showed that the product obtained by self-condensation of furfural did not have bonding strength. Then, urea and furfural could react to form urea-formaldehyde resin (UF), but the bonding strength of urea-formaldehyde resin to wood was much less than that of the lignin-based composite adhesive. Moreover, it was most surprising that the DES and furfural could not react to form urea-formaldehyde resin, and the final product was a slurry-like substance with almost no bonding strength. We believed that it was the hydrogen bond formed between choline chloride and urea that prevented the reaction of furfural with urea. This also demonstrated that no urea-formaldehyde resin was generated in the preparation of the lignin-based composite adhesive. However, in the reaction of DES + lignin (D-L), the lignin adhesive obtained had a much lower bonding strength to wood than the lignin-based composite adhesive. Finally, the DES had little or no bonding strength to wood. Therefore, this control experiment can show that the adhesive prepared in this study was a kind of composite product formed by the reaction of the DES, lignin and furfural, so we named it a lignin-based composite adhesive.

As can be seen from Fig. S3,† the characteristic peaks in the IR spectra of furfural self-condensation were still present and had not disappeared in the IR spectra of the DES-lignin adhesive. This indicated that furfural did not react with urea in the DES, which was also consistent with the experimental phenomenon we observed. While in the IR spectra of the DLF adhesive, the characteristic peak of F–F disappeared, and we could assume that furfural reacted with lignin.

Fig. 2h shows the possible mechanism of the depolymerization of lignin by the DES and the route for the synthesis of the adhesive. During alkaline chloride-based DES pretreatment, two reactions usually occur: (1) partial β–O–4′ bond breakage; (2) demethoxylation of lignin units. A similar degradation mechanism of lignin in an alkaline ChCl-based DES has also been reported, and the ChCl/urea pretreatment of lignin could oxidize the OH groups of the side chains in lignin.[50]

In order to determine the changes in the lignin structure before and after depolymerization, we performed ^{1}H NMR characterization of alkali lignin (A-L) as well as depolymerized lignin (D-L) (Fig. S13†).

Penetration of the DLF adhesive to wood

The DES can permeate the structure of wood, and dissolve and modify lignin *in situ* to remold the wood structure, indicating that the DLF adhesive can better penetrate the interior of the wood panel and enhance the adhesive strength. Fig. 3a shows a photographic image of the sample after the adhesion performance test. Compared with the original paulownia plate, the lignin-based composite adhesive with a reaction time of 6 h corroded the board most severely (Fig. 3b and c), indicating the high remolding of the wood structure therein. The lignin-based composite adhesive with a reaction time of 9 h had relatively light corrosion on the wood boards (Fig. 3d). The wood boards corroded by the lignin-based composite adhesive with a reaction time of 12 h and a reaction time of 15 h had little difference in corrosion compared with the original paulownia plate under SEM (Fig. 3b, e, and f), indicating that there was less corrosion on the wood boards, which may be the effect of the viscosity of the adhesive on the penetration behavior and the higher viscosity prevented the penetration of the DES and adhesive to the inside of the wood.

The role of the DES in the penetration of DLF-6 into the wood was further investigated using SEM (hot-pressed samples, 0.13 MPa). It can be seen that the area without DLF-6 had a large number of unfilled holes while the holes in the area containing DLF-6 were filled, indicating that the DLF-6 adhesive had penetrated into the wood (Fig. 3g–j). Then, we cut a sample along the cross-section and observed the penetration of DLF-6 with a microscope as well as a fluorescence microscope (Fig. 3k). DLF-6 had spread to both sides along the contact surface of the two pieces of wood, indicating efficient penetration (Fig. 3j). The fluorescence microscopy study also showed that there was strong fluorescence at the

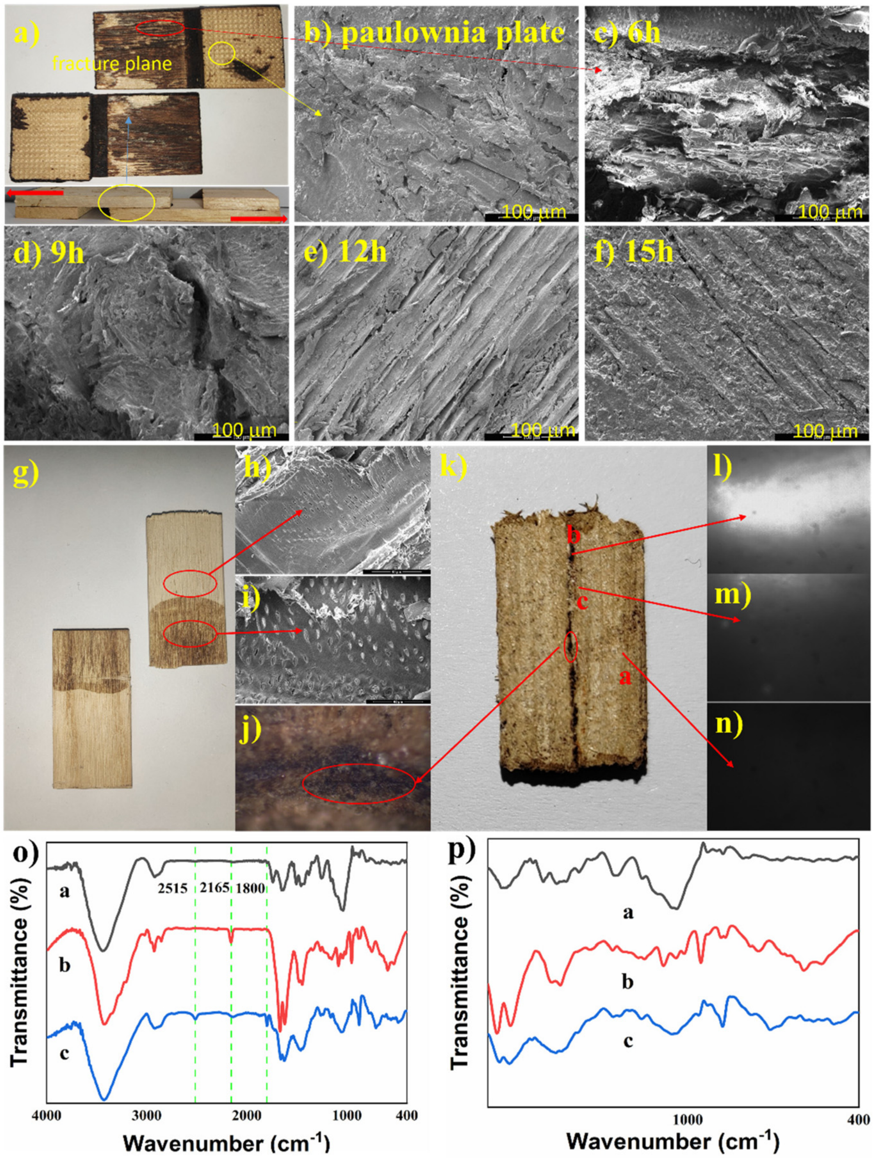


Fig. 3 (a) Pictures of samples after testing; (b) SEM image of the original paulownia plate; (c) SEM image of the degree of corrosion of the wood board by the lignin-based composite adhesive with a reaction time of 6 h. (g–j) SEM characterization of the wood penetration results of the DLF-6 adhesive; (k) microscopic characterization; (l–n) fluorescence microscopic characterization; (o and p) infrared spectroscopic characterization.

junction of the two pieces of wood (Fig. 3l). The intensity of the fluorescence underneath the junction diminished, but it still fluoresced (Fig. 3m) while the wood itself was not fluorescent (Fig. 3n). The fluorescence was emitted by DLF-6 and it also proved that it had penetrated into the wood. Finally, according to the FT-IR spectra of wood (a), DLF-6 (b), and the wood penetrated by DLF-6 (c), it can be seen that curve *c* contains almost all the characteristic peaks of curves *a* and *b*, also indicating that the position c contains both DLF-6 and wood. Moreover, new characteristic peaks appeared in curve *c*, and DLF-6 may have chemically reacted with the interior of the wood during the process of hot pressing, allowing new chemical bonds to be formed (Fig. 3o and p). Raman spectra of the three positions (a, b, and c) were also recorded. It can be seen that curves *b* and *c* had very similar characteristic peaks (Fig. S5†), which also indicated that DLF-6 had penetrated into the interior of the wood.

Not only that, we also characterized the shear specimens with Fourier transform infrared spectroscopy microscopic images in order to demonstrate the penetration effect of the adhesive into the wood (the FTIR image was recorded at 2165 cm^{-1}, a characteristic absorption peak of the DES) (Fig. S14†).

Thermal stability and flame retardance performance

As shown in Fig. S3,† the thermal stability of the adhesive increased with an increasing reaction time of 0, 6, and 15 h. From the derivative thermogravimetric (DTG) analysis curve, it can be found that the fastest decomposition rate of the adhesive for 6 h of reaction was at about 220 °C (Fig. S4†). The DES used here was rich in flame-retardant elements and was inherently non-combustible, and it provided DLF adhesives with excellent flame-retardance properties. The paulownia plate coated with the DLF-6 adhesive had passed the vertical combustion test and the fire rating reached V0 (Fig. 4a–d, Fig. S6 and Video S1†). The original paulownia plate did not pass the vertical combustion test (Video S2†). In addition, the ultimate oxygen index of the original paulownia plate was only 22%, while the paulownia plate coated with DLF-6 had an ultimate oxygen index of 30.2% (Fig. 4e).

In the vertical burning test, it was found that the surface of the DLF-6 adhesive would carbonize rapidly when it was burned at high temperatures, forming a protective layer to protect the adhesive and the board below from being burned. In order to investigate the flame retardancy of the DLF-6 adhesive, we took the DLF-6 that had been carbonized on the surface of the sample after the vertical burning test, the DFL-6 under the carbonized layer and the DLF-6 that had not been burned, and performed SEM tests on them (Fig. 4f). The carbonized layer of the DLF-6 adhesive was found to be honeycomb-like, which can prevent the flame from continuing to spread deeper where carbonization occurred (Fig. 4g–i). In contrast, the DLF-6 adhesive under the carbonized layer did not show similar holes, and its surface was flat and did not show signs of being burned (Fig. 4j–l). Moreover, it essentially had the same morphology as the unburned DLF-6 adhesive (Fig. 4m–o), with the same flat surface and no holes. Thus, it could protect the DLF-6 adhesive under its carbonized layer and the wood board under the DLF-6 adhesive. The excellent flame retardance properties made it perfectly suitable for use as a fireproof coating.

Thermal insulation performance test

The thermal diffusion coefficient was tested with a laser thermal conductivity meter. The heat diffusion coefficient of the currently reported phenolic resin fireproof coating is about 0.35 $mm^2\ s^{-1}$,[51] while the heat diffusion coefficient of DLF-6 is only 0.103 $mm^2\ s^{-1}$ (Fig. S7†). It revealed that DLF-6 adhesives had a slower temperature transfer rate and took longer to transfer temperature from one side to the other. The thermal conductivity of the DLF-6 adhesive was 0.225 W m^{-1} K^{-1}. These data showed that the DLF-6 adhesive had excellent heat

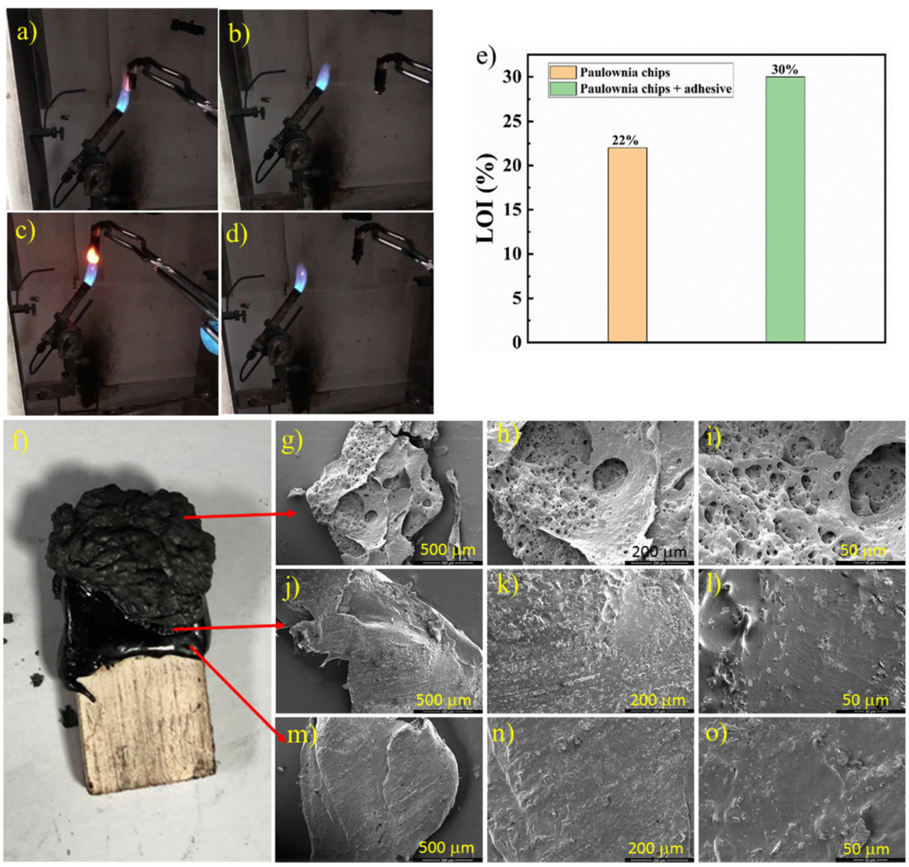


Fig. 4 Vertical combustion test figure (a–d) and limiting oxygen index (e). (a) First combustion figure; (b) first flame extinction figure; (c) second combustion figure; (d) second flame extinction figure; (e) samples tested by vertical combustion; (f–h) SEM images of the DLF-6 adhesive that had been carbonized; (i–k) SEM images of the DLF-6 adhesive under the carbonized layer; (l–o) SEM images of the DLF-6 adhesive that had not been burned at high temperature.

insulation performance and had potential as a heat insulation material (Fig. S8 and S9†).

Photothermal properties

The adhesives also showed excellent photothermal performances, which can be found in the ESI (Fig. S10 and S11† with relative discussion).

Adhesion properties to wood chips

We speculated whether the DLF-6 adhesive would also have bonding properties to wood cross-sections and wood chips. The cross-section bonding experiments of wood showed that the bonding strength of the DLF-6 adhesive to wood cross-sections was still very good, reaching 4.18 MPa, which was slightly lower than the 4.54 MPa of the wood surface (Fig. 5a–c). To study the adhesion performance of the DLF-6 adhesive to wood chips, pieces of wood were mixed with DLF-6 and hot pressed to obtain boards with a thickness of 1 mm (DLF-balsa wood) and compared with the tensile strength of balsa wood. It was found that the tensile strength of balsa wood was only 1.73 MPa while the tensile strength of DLF-balsa wood was 3.61 MPa, indicating that DLF-6 enhanced the tensile strength of balsa wood (Fig. 5d–g). These experimental results indicated that the DLF-6 adhesive had an excellent bonding effect on wood cross-sections and wood chips, which makes it more promising for a wide application.

Cost of adhesives

The cost of raw materials was an important factor when trying to achieve sustainability (Fig. S12†). The current information shows that the raw material cost of urea-furfural adhesive (UF) is 850 USD per ton, phenolic resin adhesive (PF) market price is 1800 USD per ton, epoxy resin adhesive (ER) manufacturing cost is 3880 USD per ton, polyurethane adhesive (PU) manufacturing cost is 3920 USD per ton, and in the recent development of the Clayton R epoxy soybean oil–malic–tannic acid adhesive, the cost of the raw materials is 5150 USD per ton.[52] Our lignin-based composite adhesive had a raw material cost of only $500 USD per ton. These data showed that our lignin-based composite adhesive will be an excellent alternative to petroleum-based adhesives.

Experimental

Materials

Alkali lignin was purchased from Shanghai Acmec Biochemical Co., Ltd. Choline chloride (98%, ChCl) and furfural (99%, purified by vacuum distillation) were purchased from Aladdin. Urea and anhydrous potassium carbonate were purchased from Sinopharm Chemical Corporation. Paulownia plates were purchased from Shandong Province. A galvanized iron sheet was purchased from Zhejiang Province.

Preparation of deep eutectic solvents (DESs)

ChCl and urea were first mixed in a molar ratio of 1 : 2, and then 2.1 wt% of K_2CO_3 was added. In brief, 5 g ChCl, 4.3 g urea, and 0.2 g K_2CO_3 were added into a 100 ml flask and heated at 80 °C for 30 min to form a colorless and transparent liquid, and then stored in a desiccator for later use.

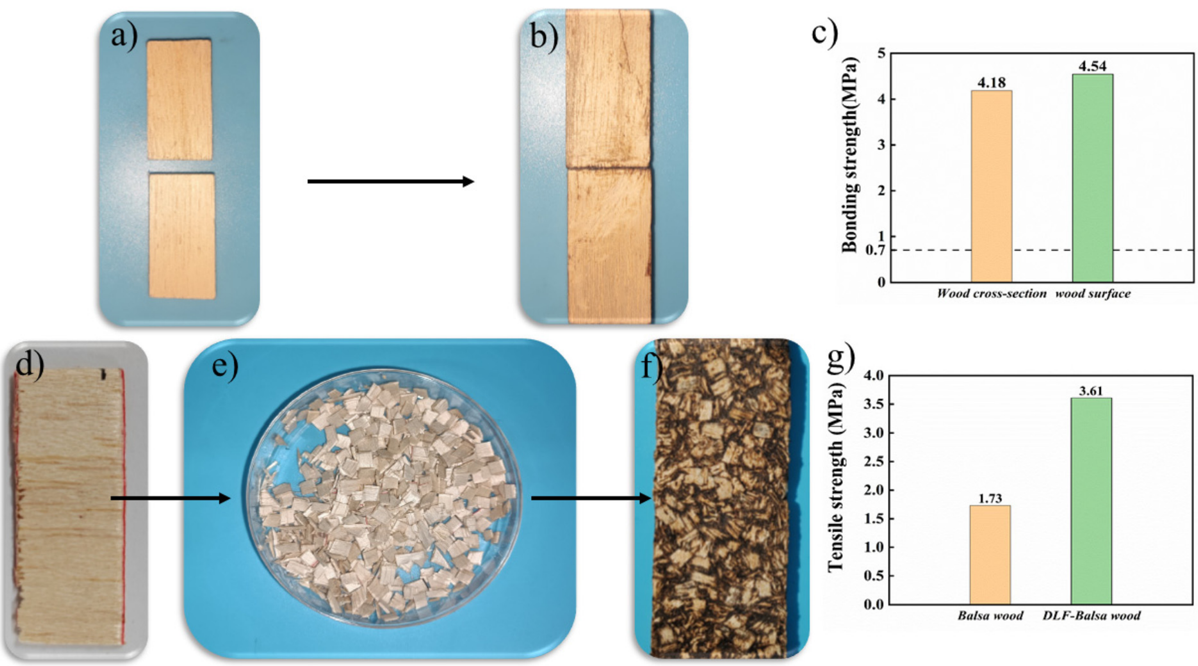


Fig. 5 (a) Paulownia plate cut along the middle; (b) paulownia plate adhered by the DLF-6 adhesive; (c) comparison of the adhesion properties of the DLF-6 adhesive to wood cross-section and wood surface; (d) balsa wood; (e) balsa wood fragments; (f) DLF-balsa wood; (g) comparison of the shear strength of balsa wood and DLF-balsa wood.

Depolymerization of lignin

5 g of alkali lignin was dispersed into the DES prepared in the previous step for depolymerization and heated at 90 °C for 12 h, which finally turned into a viscous liquid.

Preparation of the lignin-based composite adhesive

3.7 g of furfural was added to the depolymerized DES-lignin and reacted at 120 °C. After 3 h, 6 h, 9 h, 12 h, and 15 h of reaction (DLF-3, 6, 9, 12, 15), one part of the adhesive was removed and applied to the wood board while it was still hot, and the other part was sealed and stored at room temperature

FTIR characterization

FTIR spectra were measured using a Bruker EQUINOX 55 Fourier transform-infrared spectrometer with the KBr disk method. The wavenumber range is 400–4000 cm^{-1} and the spectrum resolution is 4 cm^{-1}.

^{1}H NMR characterization

^{1}H NMR (400 Hz, d-DMSO, 25 °C, TMS) spectra were acquired by dissolving 10 mg of sample in 0.5 mL of deuterated dimethyl sulfoxide and shaking to dissolve it.

Raman characterization

The samples were characterized by Raman spectroscopy using a RAMAMLOG 6 (Spex, USA) with an excitation wavelength of 325 nm.

Fluorescence microscope

NIR-II fluorescence microscopy imaging was performed based on an upright fluorescence microscope (BX51WI, Olympus) equipped with a C-Red2, 808 nm laser, and a 1000 nm LP filter.

Dynamic light scattering test

Dynamic light scattering (DLS) of alkaline lignin before (A-L) and after (D-L) being depolymerized was performed with a Malvern Zetasizer NanoZS90.

Adhesion performance at room temperature

The bonding strength of the DLF adhesive was tested following the Chinese national standard GB/T 33333-2016. The samples were prepared according to the double-layer shear specimen and triple-layer shear specimen in GB/T 33333-2016. The samples were hot-pressed for 10 min at 130 °C, 0.13 MPa, and 130 °C, 1.6 MPa, respectively. Then, they were cold-pressed under the same pressure for 12 h at room temperature. The adhesion properties were tested using the universal testing machine (MTS809) with a 200 000 N load cell at a shear rate of 5 mm min^{-1}. The adhesion strength was recorded as the maximum tensile strength.

Adhesion performance at low temperature

To test the adhesion performance of the lignin-based composite adhesive at low temperatures, the samples after hot pressing at 130 °C and 0.13 MPa were placed in an environment of −30 °C for 12 h, and then the adhesion properties were tested using the universal testing machine (MTS809).

Rheological test

The depolymerized lignin (D-L) using the DES and DLF-3, 6, 9, 12, and 15 were tested for viscosity and modulus at 130 °C using a rheometer (from TA, USA). Steady-state flow tests were performed using 25 mm plates at shear rates ranging from 0.01 to 100 rad s^{-1}.

Thermal stability testing

Thermogravimetric analysis (TGA) and derivative thermogravimetric (DTG) analysis of the lignin-based composite adhesive with reactions of 0 h, 6 h, and 15 h were performed using a NETZSCH TG 209 F1 Libra. The test temperature was from 25 to 800 °C.

Flame retardance performance

A paulownia plate was coated with a layer of the resin adhesive of 1 mm thickness and subjected to the vertical combustion test and ultimate oxygen index test, and another paulownia plate, which was not coated with this adhesive, was subjected to the same test as a blank control.

Scanning electron microscopy (SEM)

SEM characterization was performed to determine the corrosion of the lignin-based composite adhesive on the wood panels. Before testing, the wooden boards were tested for adhesion properties at room temperature (hot pressing at 0.13 MPa, 130 °C for 10 min, then cold pressed for 12 h), and the untreated paulownia plate was soaked in water to wash away the adhesive on the surface of the boards, and then freeze-dried. After the blister board coated with the lignin-based composite adhesive passed the vertical burning test, the resin on the burning surface, the resin under the burning surface, and the unburned resin were taken for SEM characterization.

Heat insulating properties

The thermal diffusion coefficient of the adhesive was tested using a laser thermal conductivity meter (NETZSCH LFA467) and its thermal conductivity was calculated. One side of the paulownia plate was coated with the adhesive, and the coated side was heated with an alcohol lamp for 10 s. The temperature of the back side was measured using an infrared thermal imager; the same control experiment was conducted using the paulownia plate without the adhesive, and the temperature of the back side was measured.

Photothermal performance

The lignin-based composite adhesive was applied to the surface of the paulownia plate, and the surface coated with the adhesive was exposed to laser irradiation of 0.92 W cm^{-2} at room temperature with an 808 nm laser; the temperature change was observed using infrared thermography (XINTEST).

Adhesion properties of adhesives to wood cross-sections and wood chips

A paulownia plate of size 3 × 15 × 40 mm was cut along the middle, and then the DES-lignin-furfural lignin-based composite adhesive (DLF adhesive) was applied to the section, after which the adhesion properties were tested by hot pressing at 130 °C and 0.2 MPa for 10 minutes and cold pressing for 12 h. The 1 mm thick balsa wood was cut into slices and then mixed with the DLF-6 adhesive so that each piece of wood was coated with DLF-6, after which it was placed in a mold, hot pressed at 130 °C and 7 MPa for 30 min and cold pressed for 2 h to test its tensile strength.

Conclusions

In this work, an alkaline DES had been used as the green solvent to prepare the lignin-based composite adhesive by a one-pot method while phenol and formaldehyde were wholly replaced by lignin and furfural, respectively. This new lignin-based composite adhesive will be an excellent alternative to phenolic and urea-formaldehyde resin adhesives. The DES-lignin-furfural lignin-based composite adhesive (DLF adhesive) had excellent bonding performance to wood, which not only exceeded the commercially available phenolic resin adhesives but also surpassed almost all the reported phenolic resin adhesives. In addition, the DLF adhesive had low-temperature resistance and showed better adhesion performance at low temperatures than at room temperature. The adhesive prepared in this study also had excellent flame retardance and thermal insulation properties and had the potential as both a fireproof material and an insulation material. The DLF adhesive had excellent adhesion properties not only to wood surfaces, but also to wood cross-sections and wood chips. The adhesive also had excellent photothermal properties and had the potential to be used as a photothermal material. In summary, this DLF adhesive can be used as a high-performance wood adhesive for low-temperature resistance and fire resistance and has the potential to be used in the future as a fireproof coating, heat insulation material, photothermal material, *etc*.

Author contributions

B. Zhan carried out the experiments and wrote the original manuscript, Z. Zhang and Y. Deng performed the data analysis, and L. Yan designed, supported and wrote the paper.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This work is supported by the National Key R&D Program of China (No. 2020YFA0710700), the National Natural Science Foundation of China (No. 52373159), and the Fundamental Research Funds for the Central Universities (No. YD2060002015).

References

1 K. Hirano and M. Asami, *React. Funct. Polym.*, 2013, **73**, 256–269.
2 N. J. L. Megson, *Nature*, 1951, **168**, 215–216.
3 R. D. Haworth, *Nature*, 1941, **147**, 255–257.
4 M. Arefmanesh, S. Nikafshar, E. R. Master and M. Nejad, *Ind. Crops Prod.*, 2022, **178**, 114604.
5 A. Bansode, M. Barde, O. Asafu-Adjaye, V. Patil, J. Hinkle, B. K. Via, S. Adhikari, A. J. Adamczyk, R. Farag, T. Elder, N. Labbé and M. L. Auad, *ACS Sustainable Chem. Eng.*, 2021, **9**, 10990–11002.
6 T. Zou, M. H. Sipponen, A. Henn and M. Osterberg, *ACS Nano*, 2021, **15**, 4811–4823.
7 S. Afewerki, X. Wang, G. U. Ruiz-Esparza, C. W. Tai, X. Kong, S. Zhou, K. Welch, P. Huang, R. Bengtsson, C. Xu and M. Stromme, *ACS Nano*, 2020, **14**, 17004–17017.
8 L. Dessbesell, M. Paleologou, M. Leitch, R. Pulkki and C. Xu, *Renewable Sustainable Energy Rev.*, 2020, **123**, 109768.
9 D. Huang, R. Li, P. Xu, T. Li, R. Deng, S. Chen and Q. Zhang, *Chem. Eng. J.*, 2020, **402**, 126237.
10 M. Turner, *Nat. Ecol. Evol.*, 2023, **7**, 1579.
11 B. L. Tardy, E. Lizundia, C. Guizani, M. Hakkarainen and M. H. Sipponen, *Mater. Today*, 2023, **65**, 122–132.
12 X. Gong, Y. Meng, J. Lu, Y. Tao, Y. Cheng and H. Wang, *Macromol. Chem. Phys.*, 2022, **223**, 2100434.
13 S. Kalami, M. Arefmanesh, E. Master and M. Nejad, *J. Appl. Polym. Sci.*, 2017, **134**, 45124.
14 C. B. T. L. Lee and T. Y. Wu, *Renewable Sustainable Energy Rev.*, 2021, **137**, 110172.
15 W. Li, H. Sun, G. Wang, W. Sui, L. Dai and C. Si, *Green Chem.*, 2023, 2241–2261.
16 Y. Zhao, K. Lu, H. Xu, L. Zhu and S. Wang, *Renewable Sustainable Energy Rev.*, 2021, **139**, 3430–3441.
17 Z. Sun, B. Fridrich, A. de Santi, S. Elangovan and K. Barta, *Chem. Rev.*, 2018, **118**, 614–678.
18 N. L. Radhika, S. Sachdeva and M. Kumar, *Fuel*, 2022, **312**, 122935.
19 M. Siahkamari, S. Emmanuel, D. B. Hodge and M. Nejad, *ACS Sustainable Chem. Eng.*, 2022, **10**, 3430–3441.
20 Z. H. Liu, B. Z. Li, J. S. Yuan and Y. J. Yuan, *Trends Biotechnol.*, 2022, **40**, 1550–1566.
21 M. Alonso, M. Oliet, F. Rodriguez, J. Garcia, M. A. Gilarranz and J. J. Rodriguez, *Bioresour. Technol.*, 2005, **96**, 1013–1018.

22 H. Paananen, L. Alvila and T. T. Pakkanen, *Sustainable Chem. Pharm.*, 2021, **20**, 100376.
23 Y. Song, Z. Wang, N. Yan, R. Zhang and J. Li, *Polymers*, 2016, **8**, 209.
24 H. Wang, T. L. Eberhardt, C. Wang, S. Gao and H. Pan, *Polymers*, 2019, **11**, 1771.
25 C. Gao, M. Li, C. Zhu, Y. Hu, T. Shen, M. Li, X. Ji, G. Lyu and W. Zhuang, *Composites, Part B*, 2021, **205**, 108530.
26 C. Zhang and F. Wang, *Acc. Chem. Res.*, 2020, **53**, 470–484.
27 V. Patil, S. Adhikari, P. Cross and H. Jahromi, *Renewable Sustainable Energy Rev.*, 2020, **133**, 110359.
28 L. Zhang, S. Zhang, X. Hu and M. Gholizadeh, *Chem. Eng. J.*, 2021, **419**, 129560.
29 C. Chio, M. Sain and W. Qin, *Renewable Sustainable Energy Rev.*, 2019, **107**, 232–249.
30 M. Alherech, S. Omolabake, C. M. Holland, G. E. Klinger, E. L. Hegg and S. S. Stahl, *ACS Cent. Sci.*, 2021, **7**, 1831–1837.
31 I. Bosque, G. Magallanes, M. Rigoulet, M. D. Karkas and C. R. J. Stephenson, *ACS Cent. Sci.*, 2017, **3**, 621–628.
32 W. Schutyser, T. Renders, S. Van den Bosch, S. F. Koelewijn, G. T. Beckham and B. F. Sels, *Chem. Soc. Rev.*, 2018, **47**, 852–908.
33 S. Nanayakkara, A. F. Patti and K. Saito, *ACS Sustainable Chem. Eng.*, 2014, **2**, 2159–2164.
34 G. F. De Gregorio, C. C. Weber, J. Gräsvik, T. Welton, A. Brandt and J. P. Hallett, *Green Chem.*, 2016, **18**, 5456–5465.
35 T. Zhang, T. Doert, H. Wang, S. Zhang and M. Ruck, *Angew. Chem., Int. Ed.*, 2021, **60**, 22148–22165.
36 H. Zhang, J. M. Vicent-Luna, S. Tao, S. Calero, R. J. J. Riobóo, M. L. Ferrer, F. del Monte and M. C. Gutiérrez, *ACS Sustainable Chem. Eng.*, 2022, **10**, 1232–1245.
37 Y. Liu, J. B. Friesen, J. B. McAlpine, D. C. Lankin, S. N. Chen and G. F. Pauli, *J. Nat. Prod.*, 2018, **81**, 679–690.
38 F. Zhang, Z. He, R. Tu, Z. Jia, Y. Wu, Y. Sun, E. Jiang and X. Xu, *ACS Sustainable Chem. Eng.*, 2020, **8**, 8562–8576.
39 F. H. B. Sosa, D. O. Abranches, A. M. da Costa Lopes, J. A. P. Coutinho and M. C. da Costa, *ACS Sustainable Chem. Eng.*, 2020, **8**, 18577–18589.
40 B. B. Hansen, S. Spittle, B. Chen, D. Poe, Y. Zhang, J. M. Klein, A. Horton, L. Adhikari, T. Zelovich, B. W. Doherty, B. Gurkan, E. J. Maginn, A. Ragauskas, M. Dadmun, T. A. Zawodzinski, G. A. Baker, M. E. Tuckerman, R. F. Savinell and J. R. Sangoro, *Chem. Rev.*, 2021, **121**, 1232–1285.
41 J. D. Mota-Morales, R. J. Sánchez-Leija, A. Carranza, J. A. Pojman, F. del Monte and G. Luna-Bárcenas, *Prog. Polym. Sci.*, 2018, **78**, 139–153.
42 E. L. Smith, A. P. Abbott and K. S. Ryder, *Chem. Rev.*, 2014, **114**, 11060–11082.
43 A. Satlewal, R. Agrawal, S. Bhagia, J. Sangoro and A. J. Ragauskas, *Biotechnol. Adv.*, 2018, **36**, 2032–2050.
44 S. Hong, H. Lian, X. Sun, D. Pan, A. Carranza, J. A. Pojman and J. D. Mota-Morales, *RSC Adv.*, 2016, **6**, 89599–89608.
45 J. Liu, J. Wang, Y. Fu and J. Chang, *RSC Adv.*, 2016, **6**, 94588–94594.
46 G. Yang, Z. Gong, X. Luo, L. Chen and L. Shuai, *Nature*, 2023, **621**, 511–515.
47 S. Y. Fu, Q.-Y. Pan, C. J. Huang, G. Yang, X.-H. Liu, L. Ye and Y. W. Mai, *Key Eng. Mater.*, 2006, **312**, 211–216.
48 F. Nakagawa and T. Sawa, *J. Adhes. Sci. Technol.*, 2001, **15**, 119–135.
49 G. Yang, B. Zheng, J.-P. Yang, G.-S. Xu and S.-Y. Fu, *J. Polym. Sci., Part A: Polym. Chem.*, 2008, **46**, 612–624.
50 H. Li, X. Li, T. You, D. Li, H. Nawaz, X. Zhang and F. Xu, *Int. J. Biol. Macromol.*, 2021, **193**, 319–327.
51 P. L. de Hoyos-Martínez, H. Issaoui, R. Herrera, J. Labidi and F. Charrier-El Bouhtoury, *ACS Sustainable Chem. Eng.*, 2021, **9**, 1729–1740.
52 C. R. Westerman, B. C. McGill and J. J. Wilker, *Nature*, 2023, **621**, 306–311.